INFORMATION TECHNOLOGY UNIVERSITY

Remote Sensing & Spatial Analytics

GIS-based School site selection in Lahore, Pakistan

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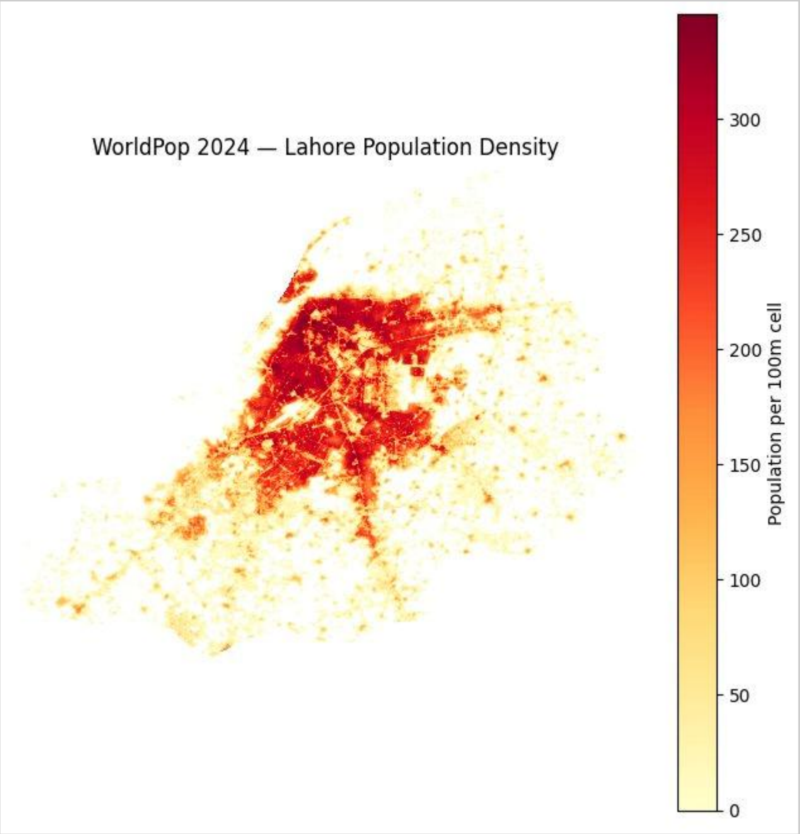
IEEE GRSS Geoscience and Remote Sensing Society LAHORE CHAPTER

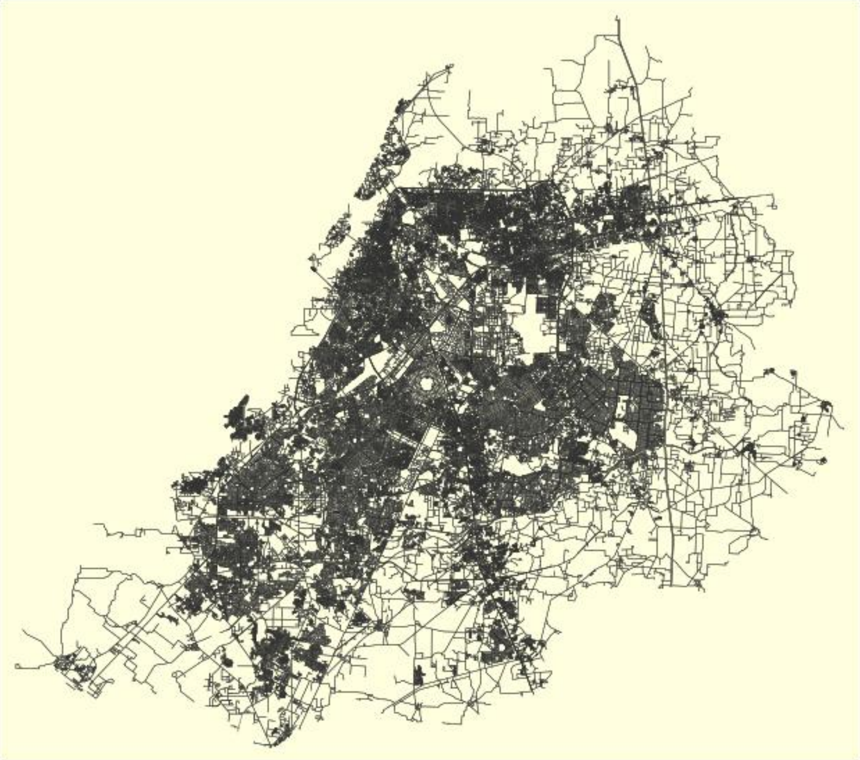
Problem Statement

- **Growing Demand:** Lahore's 14M+ population strains public schools, creating surging demand for private education — unevenly spread across the city.
- **Investment Risk:** Investors lack data on where demand is high, competition is low, and costs are realistic.
- **The Consequence:** Wrong location = low enrollment, stiff competition, early closure — all preventable with spatial evidence.



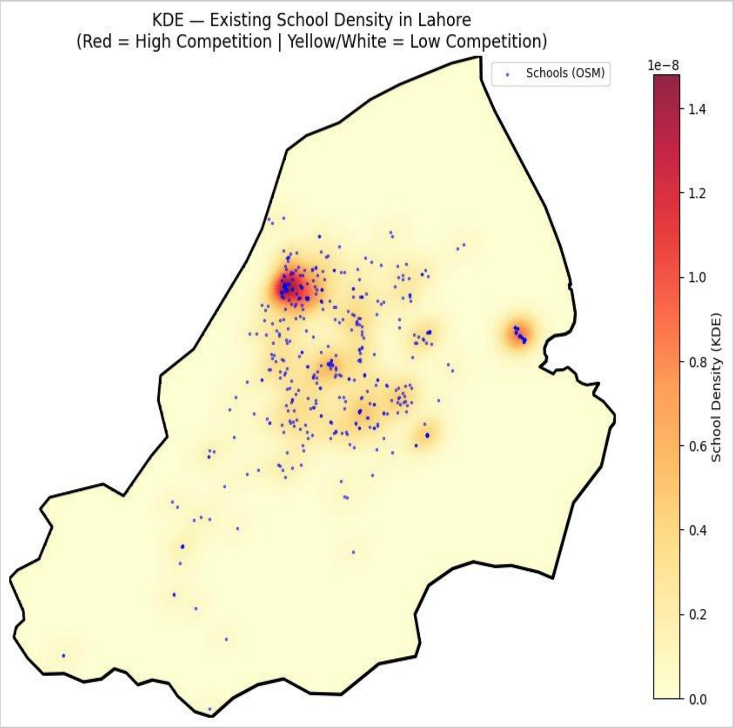
Methodology

- **Step 1 — Demand Mapping:** WorldPop + KDE (1–2 km bandwidth) on OSMnx schools identifies underserved zones with unmet child demand.
- **Step 2 — MCDA Scoring:** OSMnx road travel-time + Zameen.com property rates fed into MCDA to rank candidate sites.
- **Step 3 — Break-Even Model:** Setup costs modelled per site; minimum enrollment to break even calculated.



Results/Expected outcomes

- **Optimal Zoning Map:** Ranked, color-coded map of Lahore's top school investment zones scored by demand gap, accessibility, and cost.
- **Financial Clarity:** Site-by-site cost table + exact break-even enrollment threshold per location.
- **Confident Decisions:** Spatial data distilled into one actionable brief — right location, right budget, right time to invest.



Thank you.

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